

CATREG Analysis of Medical Charges Data

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Introduction

This study uses categorical regression with optimal scaling (CATREG) to examine factors affecting individual medical costs. The predictors include demographic and behavioural characteristics: *age of primary beneficiary*, *body mass index (BMI)*, *number of children*, *sex*, *smoking status*, and *residential region*. The outcome variable is the total *individual medical charges* (plus 1 to avoid zeros).

A preliminary inspection suggested that:

- **Age** and **BMI** have wide numeric ranges (64 and 48 distinct values) and can be treated as continuous or ordinal.
- **Smoker**, **Sex**, and **Region** are nominal categorical variables.
- **Number of children** is discrete but ordered, so it can be defined as ordinal.

But, when a variable is defined as numeric in CATREG, it is not transformed; thus, defining ordinal and nominal predictors allows nonlinearity to be modelled through optimal spacing. It is unnecessary to transform the outcome, when all predictors are being transformed to capture the non-linearity. Hence, the dependent variable (charges) which is measured on a ratio scale, is defined as **numeric**, while predictors were assigned as follows:

Variable	Scale Type in CATREG
Age and BMI	Ordinal
Smoker, Sex, Region	Nominal
Number of Children	Ordinal
Charges (Outcome)	Numeric (Linear)

This configuration allows the model to capture nonlinear effects of predictors while keeping computational intensity moderate.

Methods

The data were analysed in SPSS v30 using:

- Analyze → Regression → Optimal Scaling (CATREG)
- Convergence criterion = 0.00001, Maximum iterations = 100
- Cross-validation enabled to estimate Expected Prediction Error (EPE)

Transformations were inspected to ensure that ordinal variables followed monotonic step-wise patterns and nominal variables produced category-specific quantifications.

Results

Descriptive Statistics

The sample (N=400) is balanced by gender and region, with 19% smokers. The average age is 39.4 years (SD=13.9).

Correlation Before and After Transformation

Correlations changed only slightly (mostly within ± 0.05), suggesting mild, stabilising transformations — fine-tuning numeric spacing for ordinal variables and re-scaling nominal categories without distorting structure.

Model Fit

This saturated model (maximum degrees of freedom) shows high explanatory power with low bias:

- $R^2 = 0.771 \rightarrow 77\%$ of the variance in medical charges is explained.
- Adjusted $R^2 = 0.754 \rightarrow$ minimal overfitting.
- Apparent Prediction Error (APE) = 0.229.

Table 1: Frequency Distribution of Key Variables

Variable	Category	Frequency	Percent	Cumulative %
Charges (plus 1)	1–10000	107	26.8	26.8
	10001–20000	192	48.0	74.8
	20001–30000	41	10.3	85.0
	30001–40000	18	4.5	89.5
	40001–50000	32	8.0	97.5
	>50000	10	2.5	100.0
Sex	Female	211	52.8	52.8
	Male	189	47.3	100.0
Smoking	No	324	81.0	81.0
	Yes	76	19.0	100.0
Children+1	1	171	42.8	42.8
	2	98	24.5	67.3
	3	66	16.5	83.8
	4	53	13.3	97.0
	5–6	12	3.0	100.0
Region	Northeast	100	25.0	25.0
	Southeast	94	23.5	48.5
	Southwest	110	27.5	76.0
	Northwest	96	24.0	100.0

Table 2: Correlations Before and After Optimal Scaling

Variable Pair	Before	After
Age–BMI	.069	.071
Age–Smoker	-.034	-.037
BMI–Smoker	.040	.072
Sex–Region	-.065	.014

Table 3: Model Summary

	Multiple R	R ²	Adjusted R ²	Apparent Prediction Error
Standardized Data	0.878	0.771	0.754	0.229

Cross-validated Expected Prediction Error (EPE) = 0.262 (SE = 0.027) confirms stable generalization.

Table 4: ANOVA Summary

	SS	df	MS	F
Regression	308.568	29	10.640	43.058***
Residual	91.432	370	0.247	
Total	400.000	399		

ANOVA

The model is highly significant ($F = 43.06$, $p < .001$), confirming that predictors jointly explain a substantial portion of outcome variance.

Standardized Coefficients

Table 5: CATREG Standardized Coefficients

Predictor	Beta	SE (Bootstrap)	df	F	Sig.
Age	0.321	0.026	13	156.72	< .001
Sex	0.014	0.018	1	0.59	.443
BMI	0.178	0.026	9	45.61	< .001
Children	0.093	0.027	2	12.03	< .001
Smoking	0.780	0.024	1	1024.19	< .001
Region	0.037	0.020	3	3.33	.020

Correlations and Tolerance

Table 6: Correlations and Tolerance

Variable	Zero-Order	Partial	Part	Importance	TOL (After)	TOL (Before)
Age	.306	.555	.319	.127	.992	.991
Sex	.082	.028	.014	.001	.989	.986
BMI	.270	.343	.174	.062	.960	.951
Children	.102	.189	.092	.012	.984	.977
Smoking	.783	.851	.773	.792	.982	.985
Region	.094	.077	.037	.005	.972	.943

Discussion

CATREG achieved a strong overall fit ($R^2 = 0.77$) with stable cross-validation results (EPE=0.26). Smoking, age, and BMI are the dominant cost drivers, while children and region have minor yet significant effects. Sex does not meaningfully influence costs. High tolerance values (about 1) indicate no multicollinearity, and transformations slightly reduced tolerance, as expected, without redundancy. Transformation plots confirmed monotonic or stepwise shapes for ordinal and dichotomous variables, validating model assumptions.

In summary: Optimal scaling successfully captured the nonlinear associations while preserving interpretability. The fitted model is reliable, generalizable, and consistent with known clinical patterns.

Appendix

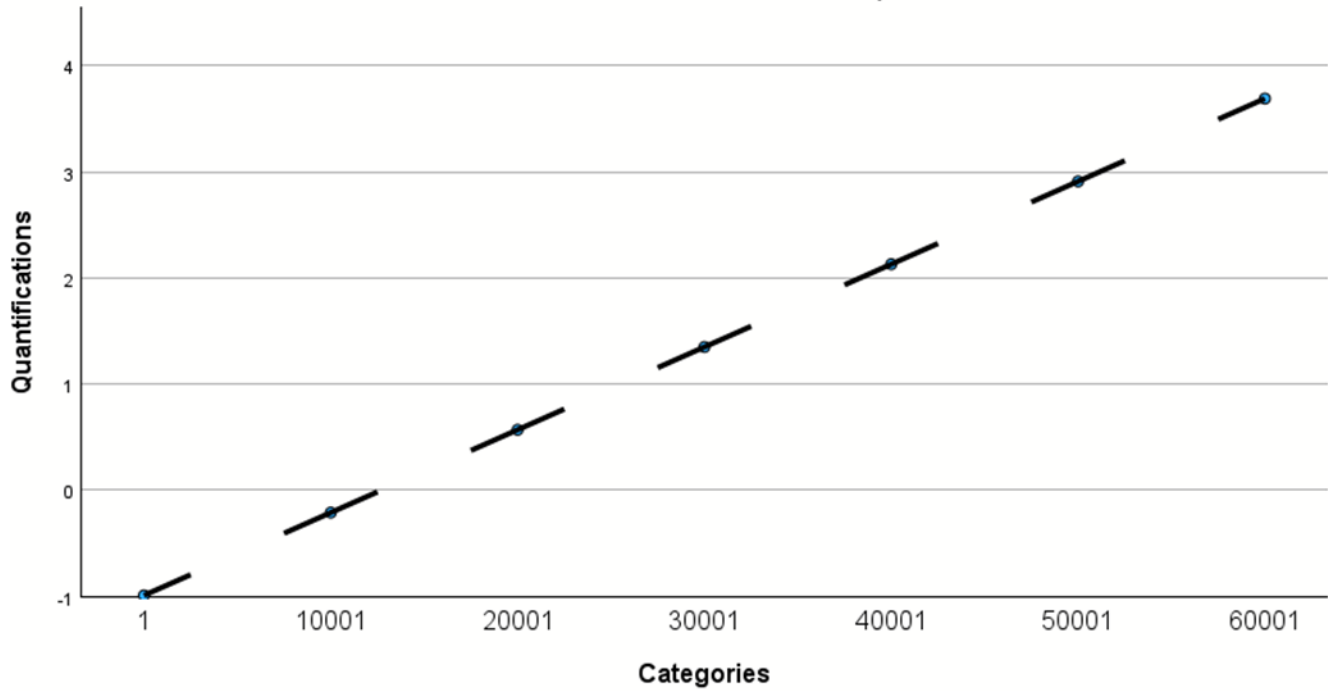
	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	age	Numeric	8	0	Age of primary beneficia...	None	None	8	Right	Scale	Input
2	sex	Numeric	8	0	insurance contractor sex...	None	None	8	Right	Nominal	Input
3	bmi	Numeric	8	0	Body mass index	None	None	8	Right	Scale	Input
4	children	Numeric	8	0	Number of children cove...	None	None	9	Right	Ordinal	Input
5	childre...	Numeric	8	0	Num of child plus 1	None	None	9	Right	Ordinal	Input
6	smoker	Numeric	8	0	Smoking: yes or no.	None	None	8	Right	Nominal	Input
7	region	Numeric	8	0	The beneficiary's reside...	None	None	8	Right	Nominal	Input
8	charges	Numeric	8	0	Individual medical costs ...	None	None	9	Right	Ordinal	Input
9	charg...	Numeric	8	0	Individual medical costs ...	None	None	9	Right	Ordinal	Target

```

3  DATASET ACTIVATE DataSet1.
4  COMPUTE charges_plus1=charges + 1.
5  EXECUTE.
6
7  COMPUTE children_plus1=children + 1.
8  EXECUTE.
9
10  FREQUENCIES VARIABLES=charges_plus1 sex children_plus1 smoker region age bmi
11  /ORDER=ANALYSIS.
12
13
14  * Chart Builder.
15  GGRAPH
16  /GRAPHDATASET NAME="graphdataset" VARIABLES=age MISSING=LISTWISE REPORTMISSING=NO
17  /GRAPHSPEC SOURCE=INLINE.
18  BEGIN GPL
19    SOURCE: s=userSource(id("graphdataset"))
20    DATA: age=col(source(s), name("age"))
21    GUIDE: axis(dim(1), label("Age of primary beneficiary."))
22    GUIDE: axis(dim(2), label("Frequency"))
23    ELEMENT: interval(position(summary.count(bin.rect(age, binCount(64)))),
24      shape.interior(shape.square))
25  END GPL.
26
27
28  CATREG VARIABLES=charges_plus1 age sex bmi children_plus1 smoker region
29  /ANALYSIS=charges_plus1(LEVEL=NUME) WITH age(LEVEL=ORDI) sex(LEVEL=NOMI) bmi(LEVEL=ORDI)
30    children_plus1(LEVEL=ORDI) smoker(LEVEL=NOMI) region(LEVEL=NOMI)
31  /MISSING=charges_plus1(LISTWISE) age(LISTWISE) sex(LISTWISE) bmi(LISTWISE)
32    children_plus1(LISTWISE) smoker(LISTWISE) region(LISTWISE)
33  /MAXITER=100
34  /CRITITER=.00001
35  /PRINT=R COEFF OCORR CORR ANOVA
36  /INITIAL=NUMERICAL
37  /PLOT=TRANS(age bmi children_plus1 sex smoker region charges_plus1)(20)
38  /REGULARIZATION=NONE
39  /RESAMPLE=CROSSVAL(10).
40

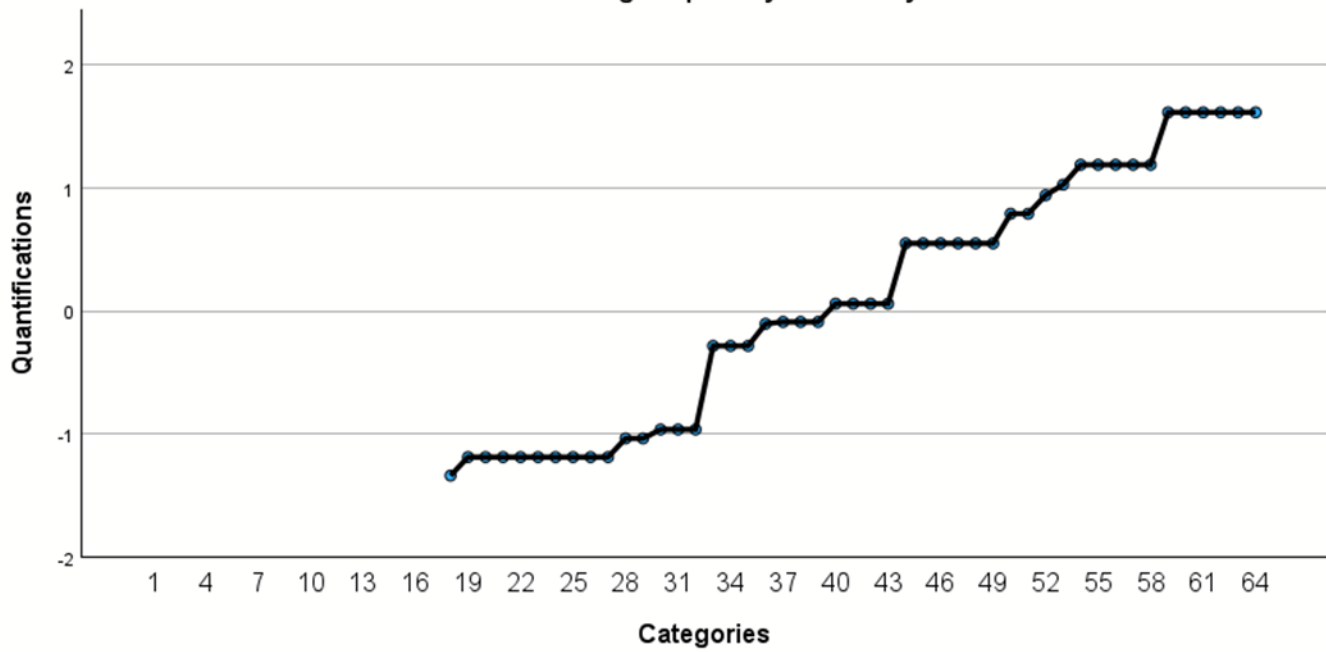
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Transformation: Individual medical costs plus 1



Optimal Scaling Level: Numerical.

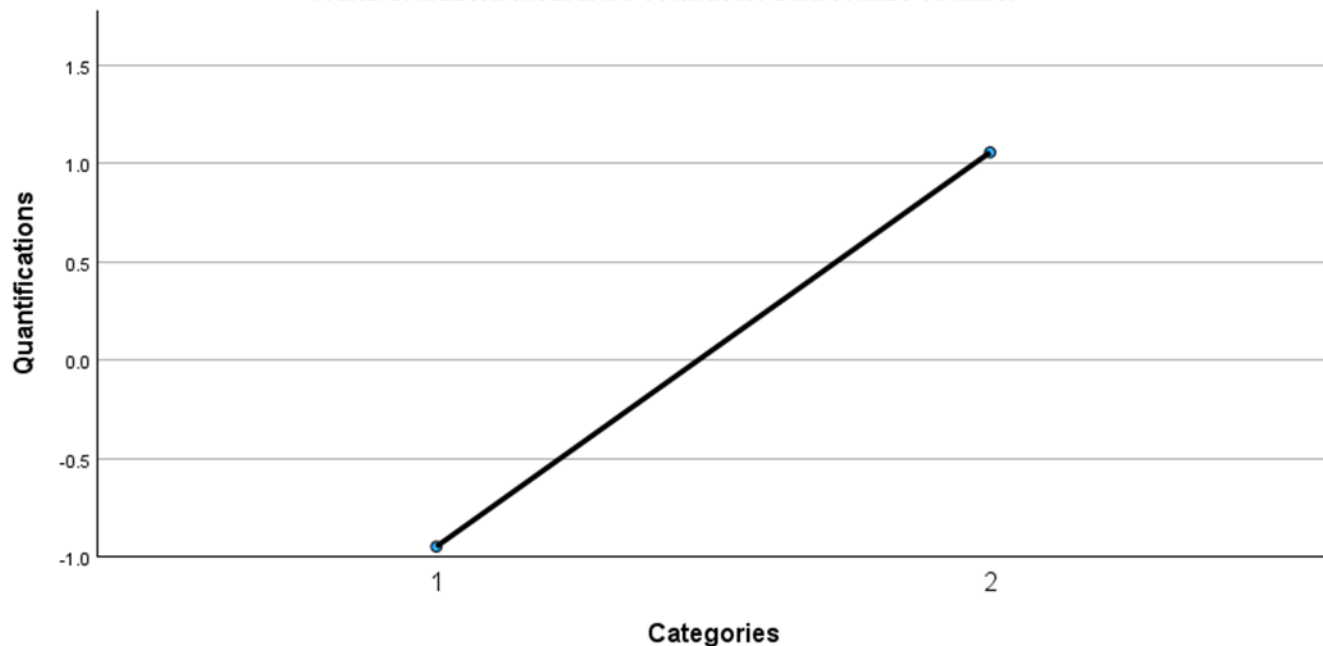
Transformation: Age of primary beneficiary.



Optimal Scaling Level: Ordinal.

Beta: .321.

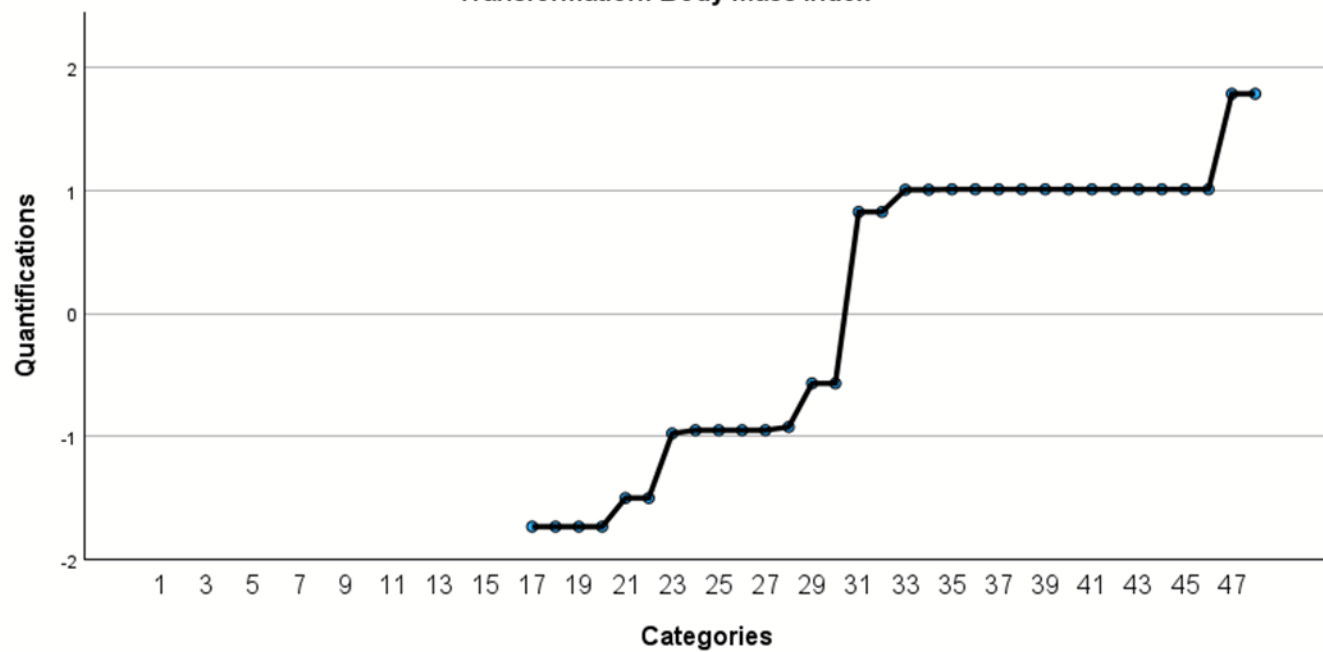
Transformation: insurance contractor sex: female or male.



Optimal Scaling Level: Nominal.

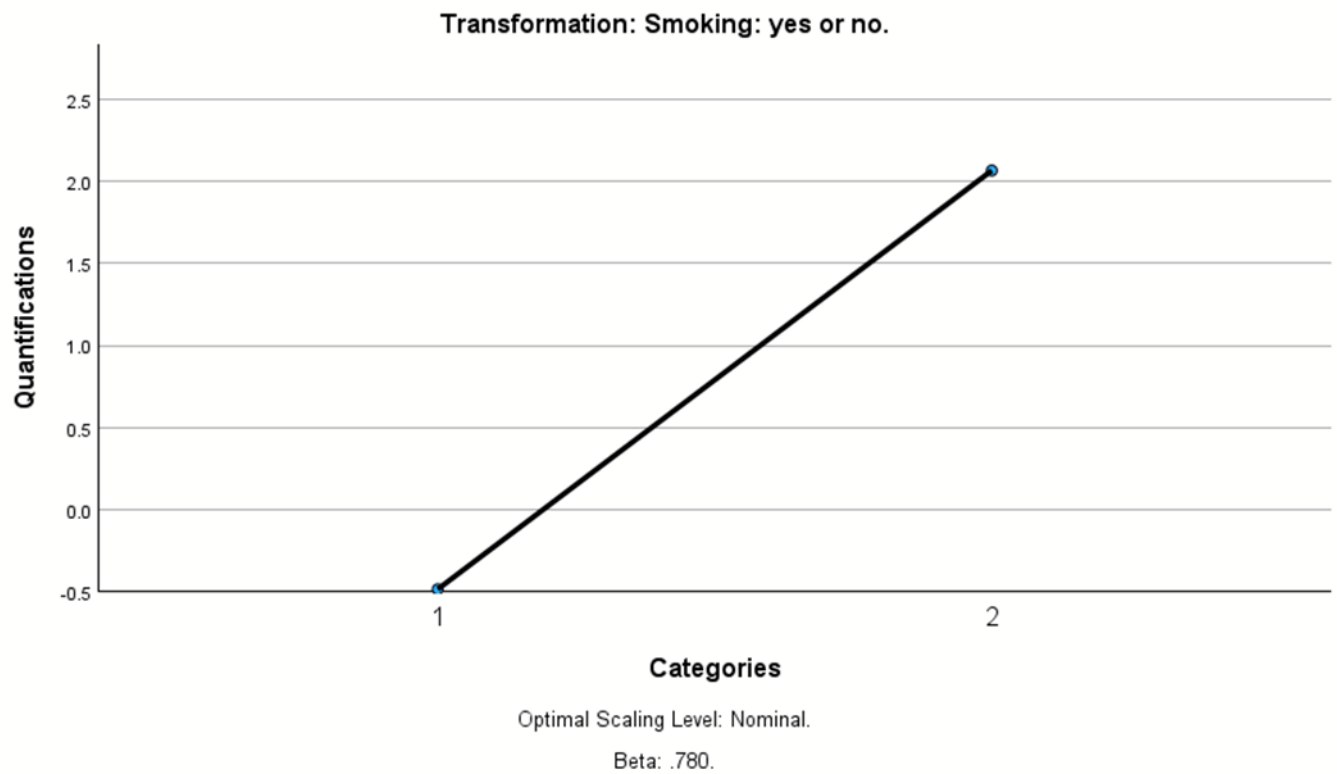
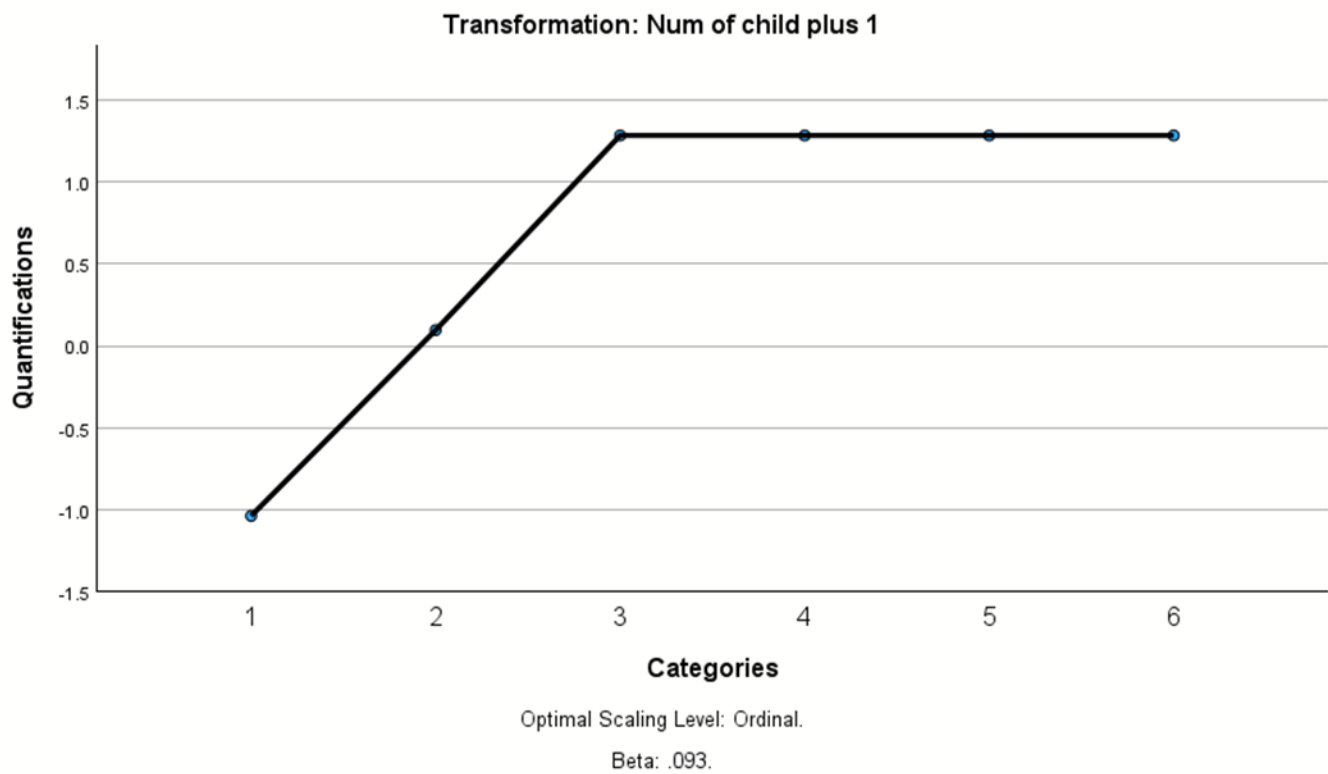
Beta: .014.

Transformation: Body mass index

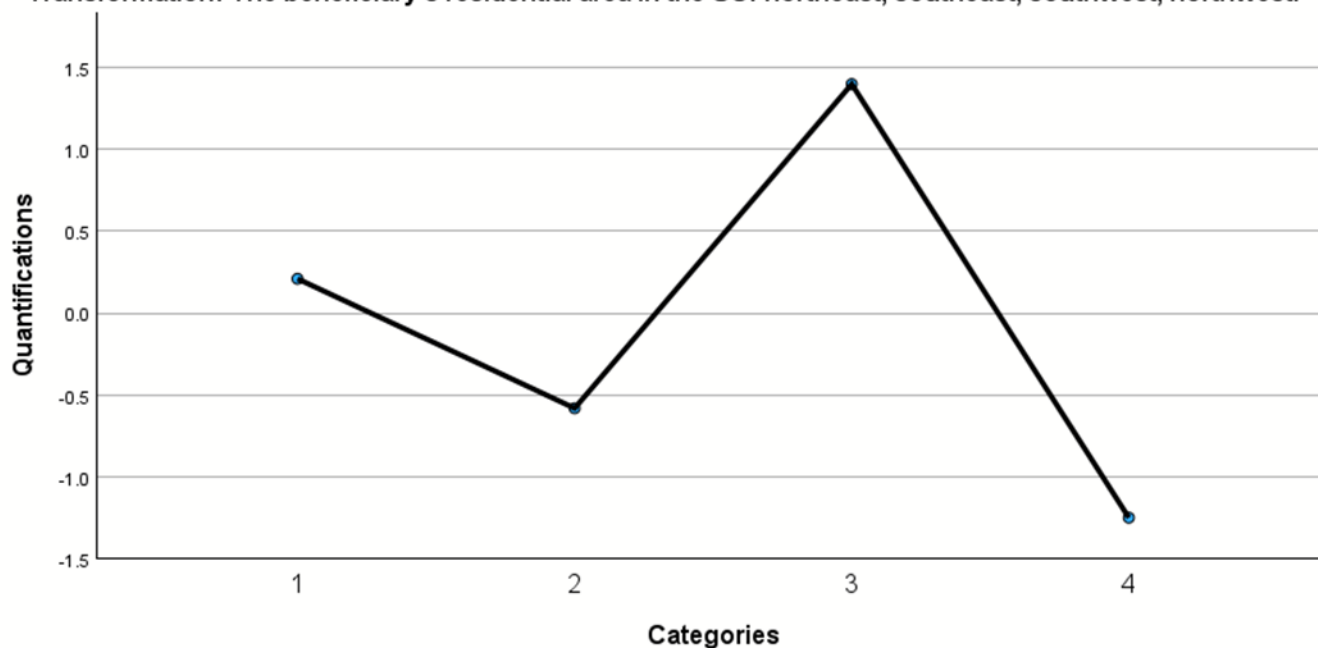


Optimal Scaling Level: Ordinal.

Beta: .178.



Transformation: The beneficiary's residential area in the US: northeast, southeast, southwest, northwest.



Optimal Scaling Level: Nominal.

Beta: .037.

Model Summary

	Multiple R	R Square	Adjusted R Square	Apparent Prediction Error	Expected Prediction Error		N ^b
					Estimate ^a	Std. Error	
Standardized Data	.878	.771	.754	.229	.262	.027	398
Raw Data				37535224.6	43056535.0	4347544.586	

Dependent Variable: Individual medical costs plus 1

Predictors: Age of primary beneficiary. insurance contractor sex: female or male. Body mass index Num of child plus 1 Smoking: yes or no. The beneficiary's residential area in the US: northeast, southeast, southwest, northwest.

a. Mean Squared Error (10 fold Cross Validation).

b. If N is smaller than the number of active (training) cases, this is due to excluding cases from estimation of the expected prediction error for reason(s) explained in the warnings table.

ANOVA

	Sum of Squares	df	Mean Square	F	Sig.
Regression	308.568	29	10.640	43.058	<.001
Residual	91.432	370	.247		
Total	400.000	399			

Dependent Variable: Individual medical costs plus 1

Predictors: Age of primary beneficiary. insurance contractor sex: female or male. Body mass index Num of child plus 1 Smoking: yes or no. The beneficiary's residential area in the US: northeast, southeast, southwest, northwest.

Coefficients

	Standardized Coefficients		df	F	Sig.
	Beta	Bootstrap (1000) Estimate of Std. Error			
Age of primary beneficiary.	.321	.026	13	156.720	<.001
insurance contractor sex: female or male.	.014	.018	1	.591	.443
Body mass index	.178	.026	9	45.610	<.001
Num of child plus 1	.093	.027	2	12.027	<.001
Smoking: yes or no.	.780	.024	1	1024.190	<.001
The beneficiary's residential area in the US: northeast, southeast, southwest, northwest.	.037	.020	3	3.334	.020

Dependent Variable: Individual medical costs plus 1

Correlations and Tolerance

	Correlations			Importance	Tolerance	
	Zero-Order	Partial	Part		After Transformation	Before Transformation
Age of primary beneficiary.	.306	.555	.319	.127	.992	.991
insurance contractor sex: female or male.	.082	.028	.014	.001	.989	.986
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